

Andrew Charabin

Toronto, Ontario

🌐 acharabin.github.io | 📞 +1 519-636-0056 | ✉ andrewcharabin@gmail.com | 📱 acharabin | 🌐 acharabin

Work Experience

Loblaw Digital

Toronto, Ontario

Data Science & Engineering Manager

Mar. 2025 – Present

- Launched an agentic service to create and manage advertising campaigns exposed through a chat interface; drove a 15% sales lift through reduced friction in the sales funnel.
- Led a team of 5 engineers accountable for machine learning solutions spanning programmatic ad serving to campaign analytics at Advance - Loblaw's media platform.
- Increased revenue by 15%+ for the sponsored search business by deploying a new click through rate (CTR) model to optimize ad placements in real-time.
- Designed a bid valuation service to recommend bid prices for ad opportunities; demonstrated a 20% increase in ROAS while maintaining media revenue.
- Enabled a new ad product to reserve display ads on Loblaw's web properties by launching an impression forecasting and booking service; generated \$15M in revenue in the first year.
- Built a simulation platform of ad placements by compressing 768-D search query and item embeddings into categories, powering campaign spend forecasts with 90% interval accuracy.

Greenstone Financial Services

Toronto, Ontario

Analytics Lead

May 2024 – Mar. 2025

- Led the Canadian analytics portfolio including business intelligence, marketing mix modelling, and agent lead allocation; recruited one Data Analyst on the team.

ADJ Industries

London, Ontario

Data Manager

Jun. 2023 – May 2024

- Enabled a business intelligence capability by implementing an on-premises MySQL database, saving \$50K+ annually in admin by centralizing data collection of payroll and customer orders.

Voices

London, Ontario

A global voice marketplace that matches and facilitates work between voice talent and their clients.

Senior Manager, Data Science & Analytics

Oct. 2021 – Jun. 2023

- Instituted an organization-wide data science function by recruiting, onboarding, and mentoring a team of 2 Data Scientists and 1 Data Analyst, creating a competitive edge for a voice marketplace.
- Created a powerful engine to drive platform retention and liquidity improvements by leading a team within the Technology department responsible for centralizing metrics, uncovering insights, and optimizing KPIs with machine learning (ML) solutions for matching, pricing, and searching.
- Reduced the time to market for new business intelligence reports and ML solutions by ~ 50% by leading a team to design a Redshift data warehouse and concomitant ELT enabled by an EC2 instance with modernized deployment, rollback, and monitoring of data models.
- Increased job hire rate by 5 points by transitioning a rules-based platform matching system to an algorithm that matches voice talent based on the likelihood of hire using an XGBoost model.
- Contributed to a 3-point improvement in the yearly client retention rate by building a platform pricing recommendation system, including an AWS lambda function to invoke SageMaker models that predict a job's price and response volume, and a CI/CD solution developed within Studio.

Manager, Analytics

Sep. 2020 - Oct. 2021

- Established a centralized analytics capability to develop and oversee a common picture of company performance through Tableau core metric reports, and surface insights from inferential

and causal models to support product and marketing strategy, helping the company achieve 30%+ YoY growth and attaining funding to integrate an ML team.

Team Lead, Analytics

Aug. 2019 - Sep. 2020

- Developed and standardized reporting and insights around platform liquidity metrics including churn, freelancer earnings, and spend retention, arming strategic discussion among leadership.

Senior Financial Analyst

Apr. 2018 – Jul. 2019

- Developed sales forecasts to guide financial targets; designed and rolled out sales comp. plans.

Skills

Languages/Tools Python, SQL (BigQuery, Redshift, SQL Server), Tableau, R, GA4, Power BI, dbt

Cloud Google (Kubernetes & Argo CD, Vertex AI, BigTable), AWS (SageMaker, Lambda)

Machine Learning Regression (spend, impressions), classification (CTR, sale), recommender systems

Analytics Experiments, causal inference, metrics & insights, visualization, data stories

MLOps & Frameworks Linux, Git, Flask, FastAPI, Cron, Airflow, XGBoost, PyTorch, TensorFlow

Soft Collegiality, structuring and communicating complexity, continuous improvement

Education

Ivey School of Business

London, Ontario

Master of Science in Management (MSc), Business Analytics

Jan. 2017 – Apr. 2018

Honors Business Administration (HBA)

Sep. 2013 – Apr. 2017

- Dean's Honor List 2014-2015

Varsity Tennis Player

Sep. 2013 – Apr. 2018

Certifications

University of Santa Cruz

Bayesian Statistics Specialization

Apr. 2024

Python Institute

Certified Associate Python Programmer

Feb. 2024

DeepLearning.AI

Natural Language Processing Specialization

Jan. 2024

University of Pennsylvania

A Crash Course in Causality: Inferring Causal Effects from Observational Data

Aug. 2023

AWS Training & Certification

Certified Machine Learning Specialty

Dec. 2022

Articles

[Goals, Principles & Constraints](#)

Mar. 2023

A Framework for Organizational Leadership

Projects

Refer to [this website](#) for an overview of prior technical projects.

[Text-To-Speech Tool with Andrew Charabin's Voice](#)

Mar. 2023

- Built upon NVIDIA's Tacotron2 implementation to create a scalable process for training models on newly recorded voices; served by a flask API.